Power BI Lab — Clean the Data & Build a Star Schema

Data Analyst batch · Wednesday 28 October 2026 · Cycle 4 · Self-guided lab

Today you will	Load 4 data files into Power BI → clean them in **Power Query** → connect them as a **star schema** → add a **Calendar** table → test with 4 visuals → upload to GitHub.
You need	Power BI Desktop (installed yesterday) and the folder **PowerBI_Lab_28Oct** with 4 files: **`Sales.csv, Products.csv, Customers.csv, Stores.csv`**
Time	About 90 minutes. Work at your own pace. Every part ends with a green **CHECK** box — do not move on until your screen matches it.
You submit	Your file **`sales_model.pbix`** + one screenshot of Model view, uploaded to your GitHub repo **`powerbi-sales-dashboard`**.

For the trainer running this class

1. Before class: copy the folder **PowerBI_Lab_28Oct** to every lab system (pen drive or shared folder), or post the zip in the class WhatsApp group.
2. Start: show this PDF on the projector for 5 minutes — read page 2 (the big picture) aloud. Then students follow the PDF on their own screens.
3. During: walk around. Most problems are in Part 2 (Power Query). Ask students who finish early to help their neighbours.
4. Use the **Answer key** on the last page to check screens. The numbers there are exact.
5. End: last 10 minutes = Part 7 (save + GitHub). Then log the class on the platform: "Power BI 3 — data model / star schema, with Power Query cleaning".

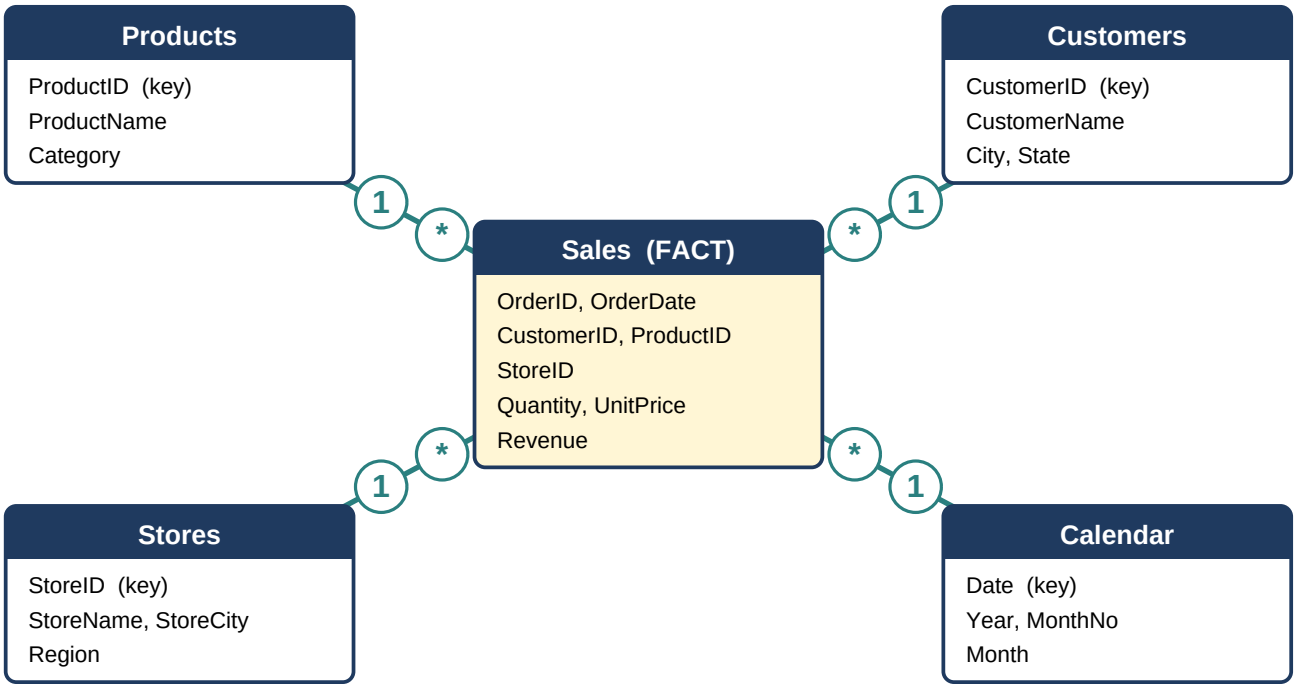
CAREFUL: If a student did not install Power BI yesterday: download **Power BI Desktop** from the Microsoft Store (free) or microsoft.com/power-bi/desktop. Pair them with a neighbour while it installs.

Part 0 — The big picture (read first, 5 min)

A shop chain in UP has sales data in **4 separate files**. Instead of one huge table, the data is split smartly:

File	What one row means	Type of table
`Sales.csv`	One order: who bought, what, where, when, how many, at what price	**FACT** table — the events / numbers
`Products.csv`	One product and its category	**DIMENSION** — describes the 'what'
`Customers.csv`	One customer and their location	**DIMENSION** — describes the 'who'
`Stores.csv`	One store and its region	**DIMENSION** — describes the 'where'

By the end you will connect them like this. The Sales table sits in the middle, and every other table connects to it. It looks like a star — so it is called a **star schema**.



WHY: A product's name is stored **once** in Products, not repeated in 300 sales rows. To answer "revenue by category" Power BI follows the line from Sales to Products. **1** = one product, ***** = many sales of that product. This is called a **one-to-many** relationship — exactly like a JOIN in SQL.

Part 1 — Load the 4 files (10 min)

1. Copy the folder **PowerBI_Lab_28Oct** to your **Desktop**.
2. Open **Power BI Desktop**. If a welcome window appears, close it. You see a blank report.
3. Click **Home** → **Get data** → **Text/CSV**.
4. Go to Desktop → PowerBI_Lab_28Oct → select **`Sales.csv`** → **Open**.
5. A preview window opens. Click **Transform Data** (NOT Load). The **Power Query Editor** opens.
6. In Power Query, click **Home** → **New Source** → **Text/CSV** and add **`Products.csv`**. Click **OK**.
7. Repeat the same for **`Customers.csv`** and **`Stores.csv`**.

CHECK: On the left side of Power Query, under **Queries**, you see 4 names: **Sales, Products, Customers, Stores**. Click each one — the data appears in the middle.

WHY: Transform Data opens Power Query — Power BI's cleaning kitchen. Raw data is almost never clean. In companies, cleaning takes more time than making charts.

Look at the right side: the **Applied Steps** panel. Every change you make is recorded here as a step — like writing SQL or pandas code, but by clicking. You can click any step to go back in time, or click the **X** to delete a wrong step.

Part 2 — Clean the data in Power Query (25 min)

Each table has problems hidden inside. Your job is to find and fix them. Do the tables in this order.

2A · Sales — remove blank rows and duplicates

8. Click **Sales** in the Queries list. Scroll down slowly — you will find some rows that are completely empty (*null* in every column).
9. Click **Home** → **Remove Rows** → **Remove Blank Rows**.
10. Now duplicates. Some orders were entered twice. Click the header of the **OrderID** column to select it.
11. Click **Home** → **Remove Rows** → **Remove Duplicates**.

CHECK: At the bottom-left of Power Query it shows the row count. Sales must now have **300 rows** (it started with 310). If it says "999+" or doesn't show, click **View** and tick **Column profile** to see counts.

2B · Sales — set the correct data types

The small icon on the left of each column header shows its type: **ABC** = text, **123** = whole number, **1.2** = decimal, a calendar icon = date.

12. Click the icon on the **OrderDate** header → choose **Date**.
13. **Quantity** → **Whole Number**. **UnitPrice** → **Decimal Number**.
14. OrderID, CustomerID, ProductID, StoreID must stay **Text (ABC)** — they are codes, not numbers to add.

CAREFUL: If Power BI asks *Change Column Type — Replace current / Add new step*, click **Replace current**.

2C · Sales — add a Revenue column

15. Click **Add Column** → **Custom Column**.
16. New column name: **Revenue**
17. Custom column formula — type exactly (or double-click the column names on the right to insert them):

```
= [Quantity] * [UnitPrice]
```

18. Click **OK**. Then click the icon on the new Revenue header → **Decimal Number**.

CHECK: The first row of Sales (ORD0223, Gel Pen Set, quantity 3, price 149) has **Revenue = 447**.

2D · Products — fix messy categories

Click **Products**. Look at the Category column: *electronics*, *ELECTRONICS*, *'Electronics '* (extra space at the end), *stationary* (wrong spelling)… To Power BI these are all **different** categories!

19. Right-click the **Category** header → **Transform** → **Trim** (removes extra spaces at start/end).
20. Right-click Category again → **Transform** → **Capitalize Each Word**.
21. Right-click Category → **Replace Values…** Value To Find: **Stationary** Replace With: **Stationery** → OK.

22. Click the small **v arrow** on the Category header to see the unique values.

CHECK: The v list shows exactly **3 categories**: Electronics, Home & Kitchen, Stationery.

WHY: Without this cleaning, a chart of "revenue by category" would show 9 bars instead of 3, and every number in it would be wrong.

2E · Customers — split Location into City and State

23. Click **Customers**. Location looks like *Varanasi, Uttar Pradesh* — two pieces of information in one column.
24. Select the Location column → **Home** → **Split Column** → **By Delimiter**.
25. Delimiter: **Comma**. Split at: **Each occurrence of the delimiter** → OK.
26. You get **Location.1** and **Location.2**. Double-click each header to rename: **City** and **State**.
27. Right-click **State** → **Transform** → **Trim** (there is a space after each comma).

CHECK: Customers has columns CustomerID, CustomerName, City, State. The first customer's State reads **Uttar Pradesh** with no space at the start.

2F · Stores — just check

Click **Stores**. This one is already clean: 5 stores, 2 regions (East UP, Central UP). Nothing to do — in real jobs, checking is also work.

2G · Load everything

28. Click **Home** → **Close & Apply**. Power BI loads all 4 cleaned tables. Wait for the loading window to finish.

CHECK: On the right side of the report, in the **Data** pane, you see 4 tables: Customers, Products, Sales, Stores.

Part 3 — Build the star schema (20 min)

29. On the far left there are 3 (or 4) icons. Hover over them. Click the one called **Model view** (it looks like 3 connected boxes).
30. You see your 4 tables as boxes. Drag them so **Sales** is in the middle and the other three are around it — like the star on page 2.
31. Power BI may have already drawn some lines automatically. Click **Home → Manage relationships** (in some versions: **Modeling → Manage relationships**). Look at the list.

You need exactly these 3 relationships now (the 4th one comes in Part 4):

From (many side *)	To (one side 1)	Cardinality	Cross filter direction
Sales [ProductID]	Products [ProductID]	Many to one (*:1)	Single
Sales [CustomerID]	Customers [CustomerID]	Many to one (*:1)	Single
Sales [StoreID]	Stores [StoreID]	Many to one (*:1)	Single

32. If a relationship in the list is **wrong** (different columns from the table above), select it and click **Delete**.
33. If one is **missing**: close the window and **drag** the column from Sales onto the same column in the other table. Example: drag **ProductID** in the Sales box and drop it on **ProductID** in the Products box.
34. Double-click any line to open it and confirm: Cardinality = **Many to one**, Cross filter = **Single**, **Make this relationship active** = ticked.

CHECK: Model view shows 3 lines, all going to Sales. Each line has a **1** at the dimension end and a ***** at the Sales end.

WHY: Relationships are Power BI's version of SQL JOINs. **Sales[ProductID] → Products[ProductID]** means the same as `Sales JOIN Products ON Sales.ProductID = Products.ProductID`. You build it once; every chart reuses it.

Part 4 — Add a Calendar (Date) table (10 min)

Companies always analyse by month, quarter and year. A proper star schema has its own **Calendar** table: one row for every day.

35. Go to **Report view** (top icon on the left). Click **Modeling → New table**. A formula bar appears at the top.
36. Delete what is there and type exactly, then press **Enter**:

```
Calendar = CALENDAR(DATE(2025, 1, 1), DATE(2025, 12, 31))
```

37. In the Data pane, click the **Calendar** table to select it. Click **Table tools → New column** (or **Modeling → New column**). Type and press Enter:

```
Year = YEAR(Calendar[Date])
```

38. Make two more columns the same way, one at a time:

```
MonthNo = MONTH(Calendar[Date])
Month = FORMAT(Calendar[Date], "MMM")
```

39. Fix the month order (otherwise Apr comes first, alphabetically): click the **Month** field in the Data pane → **Column tools → Sort by column → MonthNo**.
40. Go to **Model view**. Drag **Date** from the Calendar box onto **OrderDate** in the Sales box. Check it is **One to many** (Calendar 1 → Sales *).

CHECK: Calendar has 365 rows and columns Date, Year, MonthNo, Month. Model view now shows **4 lines** into Sales — your star is complete. Compare with the picture on page 2.

WHY: CALENDAR() is a DAX function — DAX is Power BI's formula language (like Excel formulas). You will learn more DAX tomorrow.

Part 5 — Test your model with 4 visuals (10 min)

A model is correct only if the numbers are correct. Go to **Report view** and build these. To make a visual: click an empty part of the page, click the visual icon in the **Visualizations** pane, then tick or drag fields from the Data pane.

#	Visual (icon in Visualizations pane)	Fields to use
1	**Card**	Sales → Revenue
2	**Clustered bar chart**	Y-axis: Products → Category · X-axis: Sales → Revenue
3	**Clustered column chart**	X-axis: Calendar → Month · Y-axis: Sales → Revenue
4	**Slicer**	Stores → Region

CHECK: Card shows **947.73K** (that is Rs 9,47,733). In the bar chart, **Home & Kitchen** is the longest bar and **Stationery** the shortest. Click **East UP** in the slicer — all visuals change and the card shows **611.49K**.

CAREFUL: If a chart shows the **same value for every bar**, a relationship is missing or wrong. Go back to Part 3.

Part 6 — Challenge: break it, then fix it (5–10 min)

41. Add one more visual: **Clustered column chart** with X-axis = Stores → **StoreName** and Y-axis = Sales → **Revenue**. You see 5 different bars.
42. Go to Model view. Click the line between **Sales** and **Stores**, press **Delete**, confirm.
43. Go back to Report view and look at the store chart. **Write down in your notebook: what happened, and why?**
44. Now fix it: recreate the relationship (drag StoreID onto StoreID). The chart is correct again.

CHECK: With the relationship deleted, all 5 stores show the **same** number: **947.73K**. After the fix, the tallest bar is Prayagraj Store and the shortest is Lucknow Store.

WHY: Without a relationship, Power BI does not know which sales belong to which store, so it shows the grand total for every store. This is the most common mistake in real dashboards — always test your model with a chart.

Extra challenge (for fast students)

Use your model to answer these. Write answers in your notebook and show the trainer:

45. Which **month** had the highest revenue? Which had the lowest?
46. Which **State** do most customers come from? (Hint: a bar chart of Customers → State with Count of CustomerID.)
47. Add a second slicer with Products → Category. What is the revenue of **Electronics in Central UP**?

Part 7 — Save and upload to GitHub (10 min)

48. **File → Save as** → save in the PowerBI_Lab_28Oct folder as **sales_model.pbix**.
49. Go to Model view and take a screenshot of your star schema: press **Windows + Shift + S**, select the area, then paste into Paint and save as **model.png**.
50. Open **github.com** and log in. Open your repo **powerbi-sales-dashboard**. If you don't have it: click **+** → **New repository**, name it powerbi-sales-dashboard, tick *Add a README file*, click **Create repository**.
51. Click **Add file** → **Upload files**. Drag in **sales_model.pbix**, **model.png** and the 4 CSV files. Commit message: *day 2: power query cleaning + star schema*. Click **Commit changes**.
52. Click README.md → pencil icon → add the lines below in your own words → Commit.

```
## Day 2 - Power Query + Data Model
- Cleaned data in Power Query: blank rows, duplicates,
  data types, trim, replace values, split column, custom column
- Built a star schema: Sales (fact) + Products, Customers,
  Stores, Calendar (dimensions), one-to-many relationships
- Calendar table with DAX: CALENDAR, YEAR, MONTH, FORMAT
```

CHECK: Your repo page shows sales_model.pbix, model.png and the README. Send the repo link in the class group.

Resume line: *"Cleaned and modelled sales data in Power BI using Power Query and a star schema with a DAX calendar table."* **Next class:** DAX measures and KPI cards — keep today's file.

Answer key — for the trainer

Every number below was calculated from the exact files in the folder. Use it to check screens quickly.

Check point	Correct answer
Sales rows: raw → after cleaning	310 → **300** (4 blank rows + 6 duplicates removed)
Categories after cleaning	**3**: Electronics, Home & Kitchen, Stationery (4 products each)
Customers by State	Uttar Pradesh 24 · Bihar 11 · Delhi 5 (40 customers)
Total Revenue (card)	**Rs 9,47,733** — card shows 947.73K
Total quantity sold	867 units
Revenue by Category	Home & Kitchen Rs 4,36,270 · Electronics Rs 4,21,777 · Stationery Rs 89,686
Revenue by Region	East UP Rs 6,11,494 (611.49K) · Central UP Rs 3,36,239 (336.24K)
Revenue by Store	Prayagraj Rs 2,08,119 · Varanasi Rs 2,06,757 · Gorakhpur Rs 1,96,618 · Kanpur Rs 1,76,111 · Lucknow Rs 1,60,128
Best / worst month	Best: **Jun** Rs 1,19,937 · Worst: **Dec** Rs 53,424
Store chart with Stores relationship deleted	Every store shows 947.73K (grand total)

Common problems and quick fixes

Problem on student's screen	Fix
Total revenue is higher than 947.73K	Duplicates or blank rows not removed. Power Query → Sales → check Applied Steps.
Category chart shows more than 3 bars	Trim / Capitalize / Replace step missing in Products.
Revenue column shows Error	Quantity or UnitPrice is still text. Change their types to numbers, then re-create Revenue.
Every bar shows the same number	Relationship missing or between wrong columns. Model view → Manage relationships.
Months in wrong order (Apr, Aug, Dec…)	Month → Column tools → Sort by column → MonthNo.
"New table" formula gives an error	Check spelling and commas: Calendar = CALENDAR(DATE(2025, 1, 1), DATE(2025, 12, 31))
Revenue in a visual shows Count, not Sum	Click the v next to Revenue in the visual's field well → choose Sum.